UNIRA

One Humanity, One Future

A testable framework for global resource cooperation, human wellbeing and sustainable production

WORKING PAPER 01 • Integrated research specification and preliminary findings • 3 November 2026

Status: preprint / conceptual research programme. Not peer reviewed. No claim of demonstrated global feasibility.

Executive summary

UNIRA proposes a democratically governed global economic coordination framework in which essential needs are guaranteed, regional production is cooperatively planned, work is normally limited to 30 hours per week, and resource use remains within ecological constraints. Countries retain local democratic and cultural autonomy. The model aims to replace geopolitical competition over essential resources with accountable multilateral cooperation.

The current evidence supports only a narrow finding: a transparent input-output demonstration can calculate inter-sector requirements and identify synthetic bottlenecks. No calibrated global feasibility finding exists yet. Political consent, allocation incentives, institutional safeguards and transition costs remain major unresolved challenges.

1. Research questions and falsifiable hypotheses

- RQ1: Can a coordinated production-and-distribution network meet specified nutrition, housing, health, education, mobility and energy service standards within physical and environmental limits?
- RQ2: Under what labour participation, productivity and sectoral skill conditions is a 30-hour standard workweek compatible with those standards?
- RQ3: Does coordinated allocation outperform credible mixed-economy reform baselines on unmet essential needs, ecological impact and shock resilience?
- RQ4: Can democratic global institutions prevent coercion, corruption, discrimination and persistent resource capture?

H1: UNIRA reduces the proportion of unmet essential needs relative to the benchmark under equal resource constraints. H2: It does not increase life-cycle emissions or material extraction. H3: It maintains or improves recovery from supply shocks. Each hypothesis must be rejected if independently reproducible results do not support it.

2. Agreed policy specification

Area	Current proposed rule	Open design issue
Work	6 hours/day, 5 days/week (30 hours)	Leave, care work, education, retirement and sectoral shift coverage
Pay	5,000 U/month base; maximum individual salary 25,000 U/month	Job differentials, non-wage incentives and inflation
Medical protection	5,000 U/month for medically unable-to-work adults	Eligibility, autonomy and disability rights
Children	Up to 1,000 U per child per month	Means-testing and household definitions
Child benefit ceiling	Child cash benefit = min(1,000×children, max(0, 15,000 − adult household cash income))	Include medical income in adult cash income; never reduce adult guarantee
Essential services	Guaranteed access to healthcare, education, food, water and adequate housing	Exact universal service basket and priority rules
Firms	Operating costs and reserves covered before distributable surplus funds public investment	Ownership, bankruptcy, innovation and capital allocation
Governance	Global democratic coordination with strong local autonomy	Treaty ratification, exit, rights and judicial review

The 15,000 U threshold limits child cash benefits only; it does not cap adult household earnings. U is a proposed accounting unit, not convertible to USD without an independently specified price basket and purchasing-power framework.

3. Accounting architecture: physical and monetary ledgers

Physical ledger: goods and services by region, sector, period and unit (kcal, tonnes, TWh, dwellings, clinician-hours, passenger-km). Monetary ledger: wages, operating costs, depreciation, capital investment, tax/contributions, public services, social benefits and reserves. Do not infer physical feasibility from balanced monetary accounts.

Static core: $x = (I - A)^{-1} y$, with A[i,j] intermediate input i per unit of gross output j, x gross output and y final demand. Feasibility also requires $x \leq$ installed production capacity, labour hours by occupation ≤ available qualified labour, and material/energy/emissions/transport limits. Regional allocation must be optimised rather than assigned by simple capacity shares.

4. Preliminary reproducible model: V2.2

The existing prototype uses 3 hypothetical regions, 4 sectors (food, energy, materials, housing) and entirely synthetic coefficients, capacity, labour and final demand. The following figures are dimensionless illustrative units, not global measurements.

Sector	Synthetic final demand	Synthetic gross production required
food	90.0	106.377
energy	65.0	113.64
materials	45.0	88.773
housing	40.0	45.336

In the baseline, intermediate industrial inputs raise gross output above final demand. In a hypothetical 70% energy-capacity loss in Region A, the aggregate energy shortage is 14.14 synthetic units. These are arithmetic properties of the illustrative model only. They do not imply real-world food, energy or housing shortages.

5. Labour feasibility: conditional arithmetic

For an illustrative 4.35 billion workers at 30 hours/week, available weekly hours = 130.5 billion. A comparison benchmark of 3.54 billion employed persons × 43 hours/week gives 152.22 billion hours/week. Matching output would require an average output-per-hour ratio of 152.22/130.5 ≈ 1.166, or about 16.6% more, assuming equal output composition and hours comparability. This is NOT an empirically established productivity requirement: worker count, hours, holidays, skills, unpaid care and sectoral structure differ.

Labour scenario sensitivities are essential. A larger workforce is not interchangeable with specialised clinical, engineering or teaching labour. A reduction in hours may require more shift workers and infrastructure investment.

6. Global dataset plan and evidence quality

Domain	Candidate authoritative source	Required fields / validation
Population	UN World Population Prospects	Age, region, fertility, household structures; year harmonisation
Labour	ILOSTAT / ILO	Employment, actual hours, occupation, age, informal work
Food	FAOSTAT / FAO food balance sheets	Calories, nutrients, crop yields, loss, trade and land
Energy	IEA / Ember / World Bank	Generation, final consumption, grid capacity, regional access
Input-output	OECD ICIO / Eurostat /	Inter-sector coefficients,

	EXIOBASE	trade flows, price-year and licence
Housing	UN-Habitat / national censuses	Households, quality, overcrowding, construction capacity
Health	WHO Global Health Observatory	Workforce, service coverage, needs, clinical capacity
Education	UNESCO UIS	Teachers, enrolment, facilities and quality
Environment	UNEP IRP / IPCC / Global Carbon Project	Material footprint, emissions, land and water stress

Data must be accompanied by source URLs, extraction date, units, year, geographic coverage, licence, missingness, uncertainty and transformation scripts. No real-world sector dataset has been downloaded or calibrated in this working paper.

7. Comparison design: do not compare against a straw man

Benchmark B0: current global economy with observed regional constraints. B1: realistic mixed-economy reform (universal basic services, improved trade, carbon policy, targeted transfers). U1: UNIRA with partial coordination. U2: full proposed coordination. All cases share population, climate, technology and resource assumptions; only institutional rules differ. Compare unmet needs, life expectancy proxies, cost, hours, emissions, inequality and crisis recovery, with uncertainty intervals.

8. Governance, incentives and human rights

A world coordination authority poses serious risks of concentrated power, surveillance, coercive allocation and suppression of dissent. The design must include two representative chambers, subsidiarity, constitutional rights, independent courts, auditability, public participation, protected privacy, freedom of movement and expression, and judicially reviewable allocation decisions. Resource-sharing agreements must respect local and Indigenous rights and ecological stewardship.

Production plans need incentive-compatible reporting: regional producers may overstate needs or understate capacity. Independent verification, distributed data governance, contestable planning assumptions, whistleblower protection and appeal mechanisms are necessary. Essential services should not depend on political loyalty or compulsory labour.

9. Monetary architecture and fiscal balance

Monthly payroll of 4.35 billion workers at the base 5,000 U would be 21.75 trillion U/month, or 261 trillion U/year before any salary differentiation. This is an employer labour-cost floor under the scenario, NOT an estimate of public spending. A hypothetical

2.4 billion children would have a 28.8 trillion U/year theoretical maximum child allowance before household means-testing. Actual benefits require household microdata.

The 5,000 U medical guarantee must be separately costed using medically eligible persons rather than assuming all non-working adults are disabled. Medical guarantees count toward adult household cash income when assessing child allowance in this proposed implementation, without reducing the medical payment itself.

The UNIRA central bank would need an explicit nominal anchor (e.g. stable essential-services basket), a policy for inflation and shortages, a payments architecture, fiscal transfers and transparent accounting. Nominal U values cannot prove affordability without quantities and prices.

10. Transition and crisis resilience

Adoption should be voluntary and treaty-based, staged through interoperable statistics, cross-border emergency reserves, common technical standards, pilot regional procurement and democratic evaluation. A single global monetary union before fiscal capacity and legitimate governance would create serious risk. Disarmament must be verified and reciprocal; disaster response and civilian protection remain necessary.

Stress tests: drought, crop disease, regional grid failure, pandemic, transport blockade, material embargo, cyberattack, labour shortages, misinformation, corruption and unexpected migration. Preserve multiple production sites, reserve inventories and emergency substitution pathways.

11. Research risks and stopping rules

- Stop or redesign any scenario that violates minimum nutrition, clinical care or basic rights.
- Reject any economic claim not supported by unit-consistent and reproducible data.
- Report negative outcomes and uncertainty; never infer that cooperation automatically removes scarcity or conflict.
- Reject policy comparisons that use different climate, population or technology assumptions.
- Submit the calibrated model to independent economists, engineers, climate scientists and governance researchers before strong feasibility claims.

12. Publication pathway and final deliverables

Stage A: publish this working paper and the transparent prototype. Stage B: obtain harmonised public datasets and implement constrained multi-regional optimisation. Stage C: produce sensitivity analysis and independent replication. Stage D: submit a peer-reviewed research manuscript. Stage E: publish the public-facing book, clearly distinguishing evidence, proposals and speculation.

A full scientific conclusion is not possible at this stage. This document is a complete conceptual research foundation and transparent preliminary demonstration, not a completed empirical proof.

Selected starting references (verify versions and access dates before publication)

UN World Population Prospects: https://population.un.org/wpp/

ILOSTAT: https://ilostat.ilo.org/data/

FAOSTAT: https://www.fao.org/faostat/en/

IEA Global Energy Review 2025: https://www.iea.org/reports/global-energy-review-2025

OECD Inter-Country Input-Output: https://www.oecd.org/en/data/datasets/inter-country-input-output-tables.html

UN-Habitat: https://unhabitat.org/

WHO Global Health Observatory: https://www.who.int/data/gho

UNESCO UIS: https://uis.unesco.org/

UNEP International Resource Panel: https://www.resourcepanel.org/

Disclosure: This document was prepared with AI assistance. Source entries are research leads, not a claim that each dataset was downloaded or independently audited. All synthetic results are labelled as such.